CS 24 AP Computer Science A Review

Week 0: Exam Strategy

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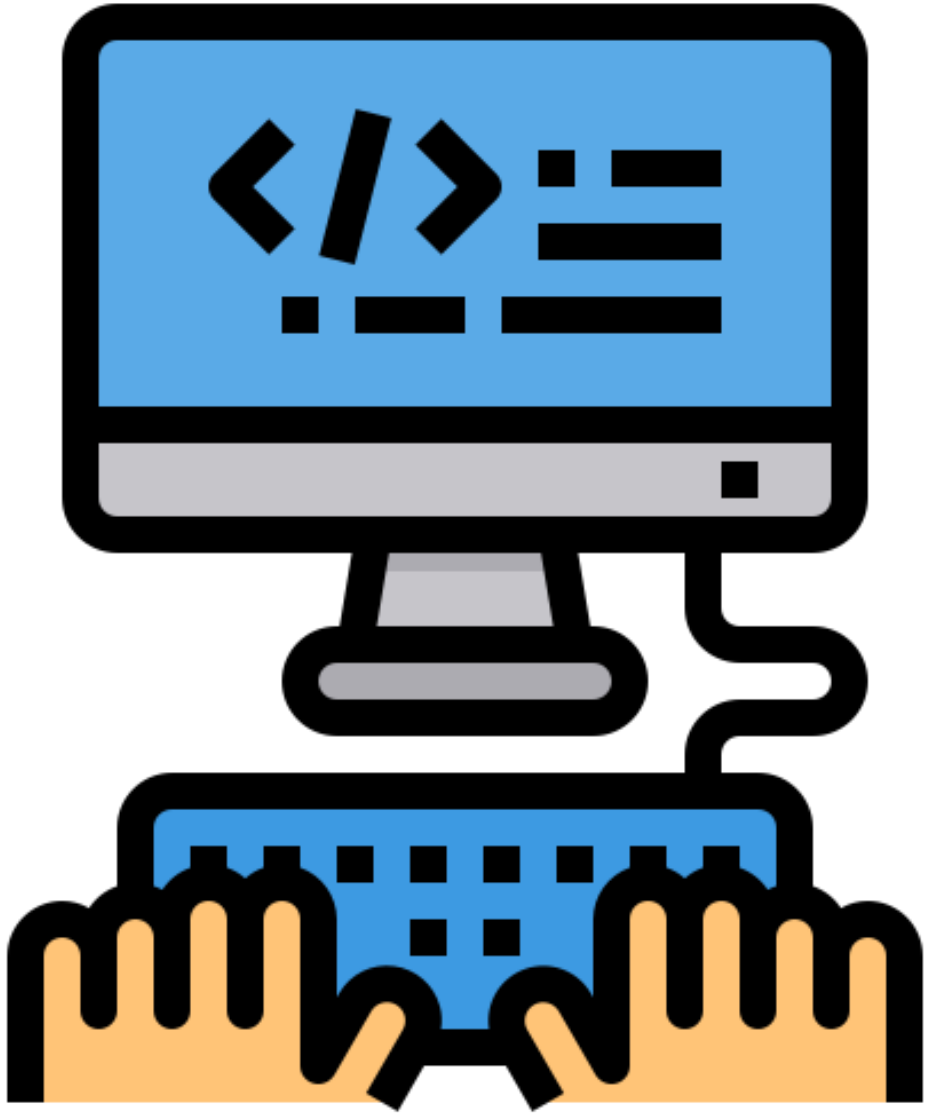
Topics

- The Structure of the AP Computer Science A Exam
- How the AP Computer Science A Exam is Scored
- Free Response Questions
- The Exam is Fully Digital
- Multiple Choice Section
- Free Response Section



The Structure of the AP Computer Science A Exam

SECTION 1

The Structure of the AP Computer Science A Exam

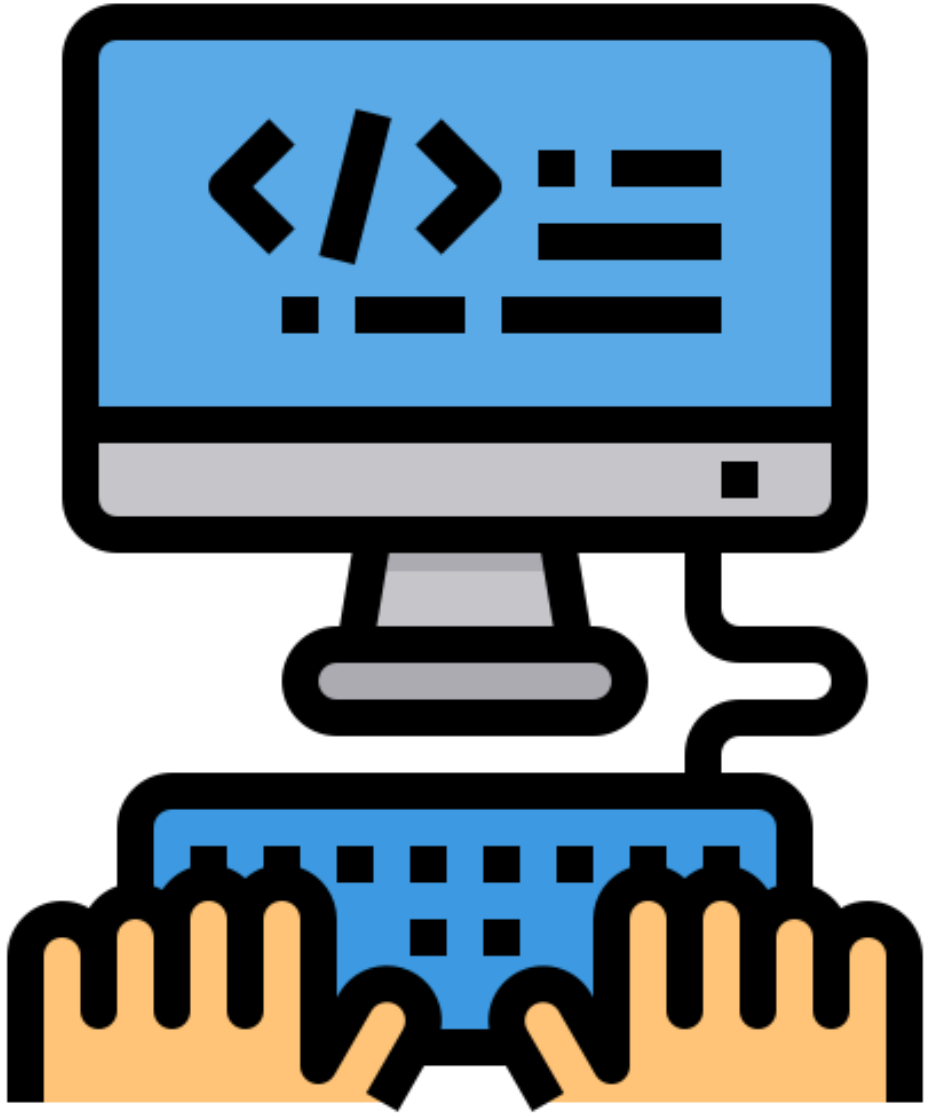
Section	Question Type	Number of Questions	Time Allowed	Percent of Final Grade
I	**Multiple Choice**	**42**	**90 min.**	**55%**
II	**Free Response** Question 1: Methods and Control Structures (7 pts.) Question 2: Class (7 pts) Question 3: Data Analysis and ArrayList (5 pts) Question 4: 2D Array	**4**	**90 min.**	**45%** 12.5% 12.6% 9% 10.8%

The Structure of the AP Computer Science A Exam

Units	Exam Weighting
Unit 1: Using Objects and Methods	15-25%
Unit 2: Selection and Iteration	25-35%
Unit 3: Class Creation	10-18%
Unit 4: Data Collections	30-40%

The Java Quick Reference

- You will have access to the Java Quick Reference for both the multiple-choice and free- response sections of the AP exam.
- The quick reference contains specifications for methods and constants from the Java library for the **Object**, **Integer**, **Double**, **String**, **Math**, and **ArrayList** classes.



How the AP Computer Science A Exam is Scored

SECTION 2

Using and Interpreting AP Scores

The extensive work done by college faculty and AP teachers in the development of the course and exam and throughout the scoring process ensures that AP Exam scores accurately represent students' achievement in the equivalent college course. Frequent and regular research studies establish the validity of AP scores as follows:

AP Score	Credit Recommendation	College Grade Equivalent
5	Extremely well qualified	A
4	Well qualified	A-, B+, B
3	Qualified	B-, C+, C
2	Possibly qualified	n/a
1	No recommendation	n/a

AP 2025 AP Exam Score Distributions

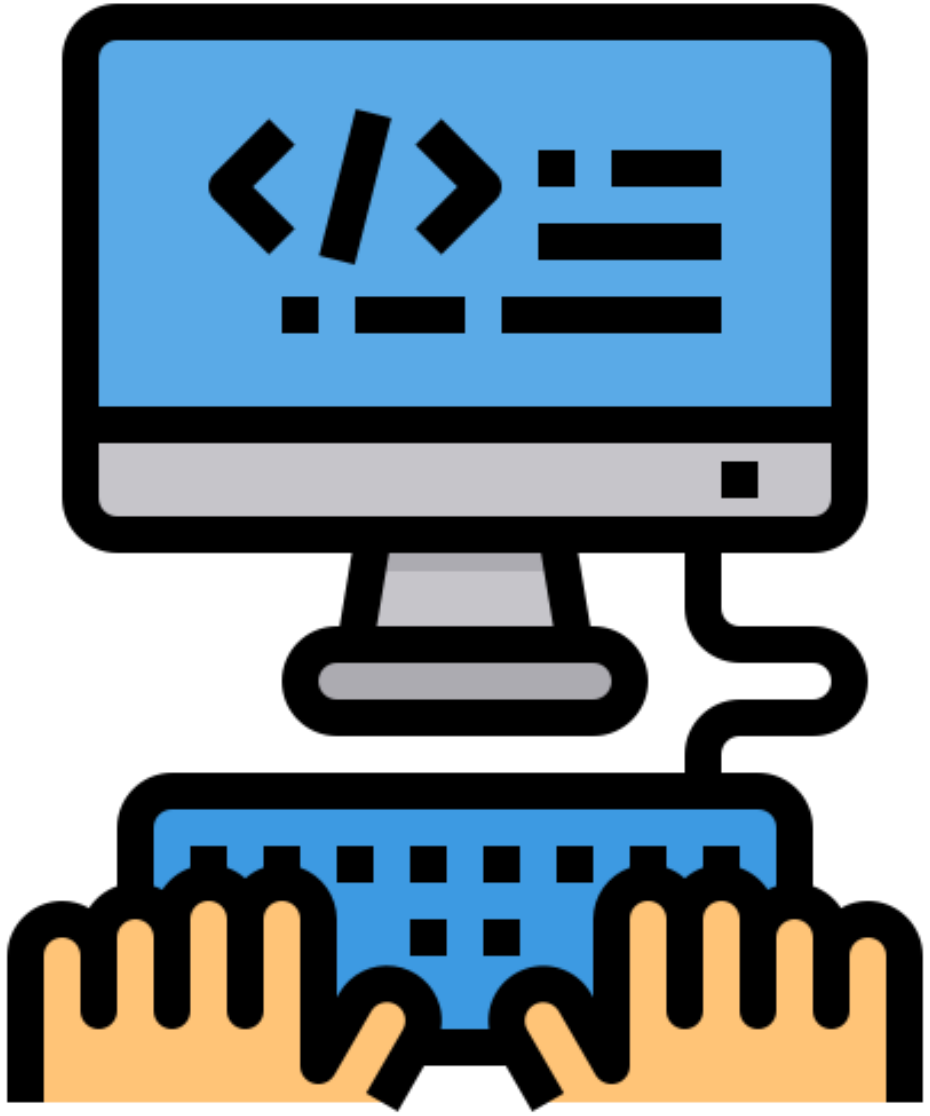
AP Computer Science A

AP Score	% of Exams
5	25%
4	22%
3	20%
2	10%
1	23%

Trevor Packer
@AP_Trevor

AP Computer Science A MCQs: strong performance on primitive types, Boolean expressions, and if statements (units 1 & 3); 44% of students earned 7-8 of these 8 points.

@AP_Trevor | @ap-trevor.bsky.social



Free Response Questions

SECTION 3

1-Point Penalty

- Extraneous code that causes a side effect or prevents earning points in the rubric (e.g. information written to output)
- Local variables used but none declared.
- Destruction of persistent data (e.g., changing value referenced by parameter)
- Void method or constructor that returns a value

No Penalty

- Extraneous code that causes no side effect
- Extraneous code that is unreachable and would not have earned points in the rubric
- Spelling/case discrepancies where there is no ambiguity
- Local variable not declared, provided that other variables are declared in some part.
- `private` qualifier on local variable.

No Penalty

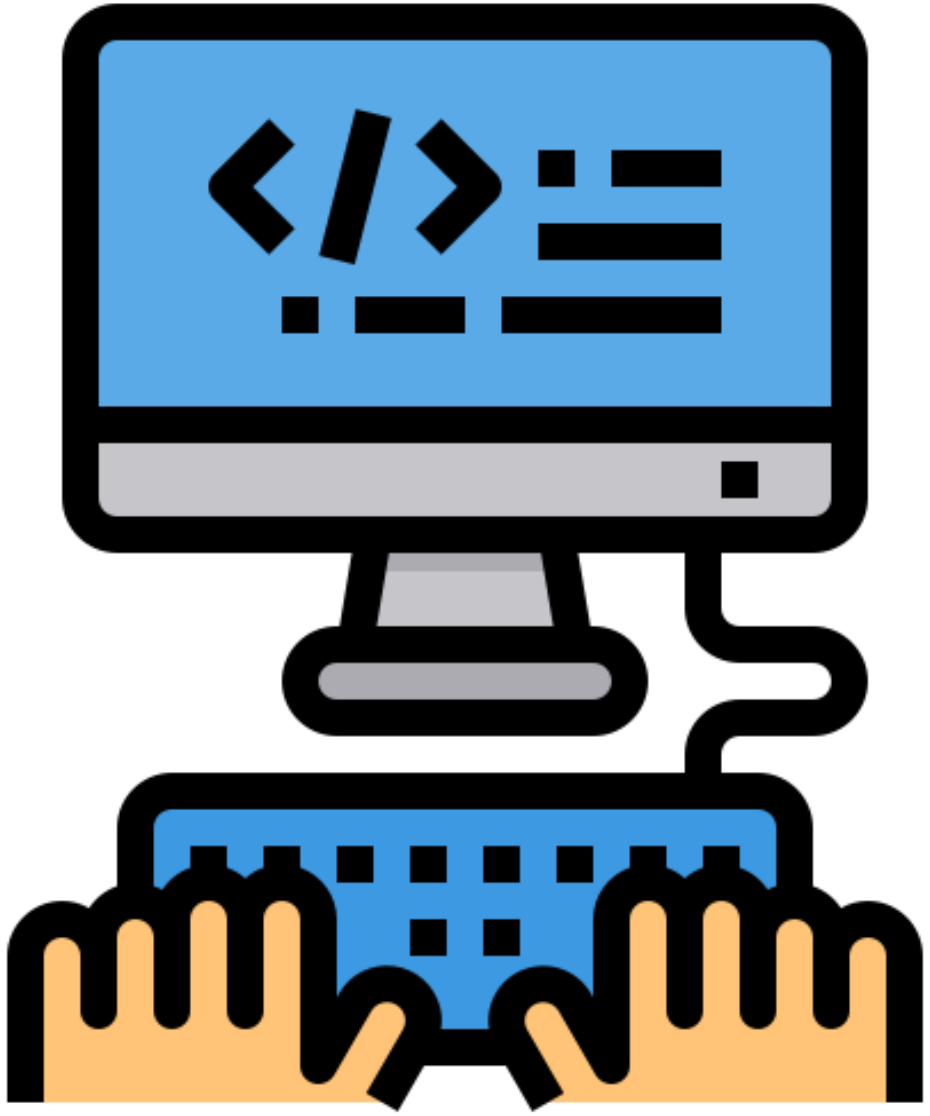
- Missing **`public`** qualifier on class or constructor header.
- Keyword used as an identifier
- Common mathematical symbols used for operators
- `[ ]` vs `( )` vs `<>`
- `=` instead of `==` (and vice versa)
- Array/Collection element access confusion (`[ ]` vs get for r-values)
- Array/Collection element modification confusion (`[ ]` vs set for l-values)

No Penalty

- Length/size confusion for array, String and ArrayList, with or without `()`
- Extraneous `[]` when referencing entire array
- `[i, j]` instead of `[i][j]`
- Extraneous size in array declaration.
- Missing; provided that line breaks and indentation clearly convey intent
- Missing `{}` where indentation clearly conveys intent and `{}` are used elsewhere.

No Penalty

- Missing （）on parameter-less method or constructor invocations
- Missing （）around if/while conditions
- Use of local variable outside declared scope (must be within same method body)
- Failure to case object retrieved from nongeneric collection



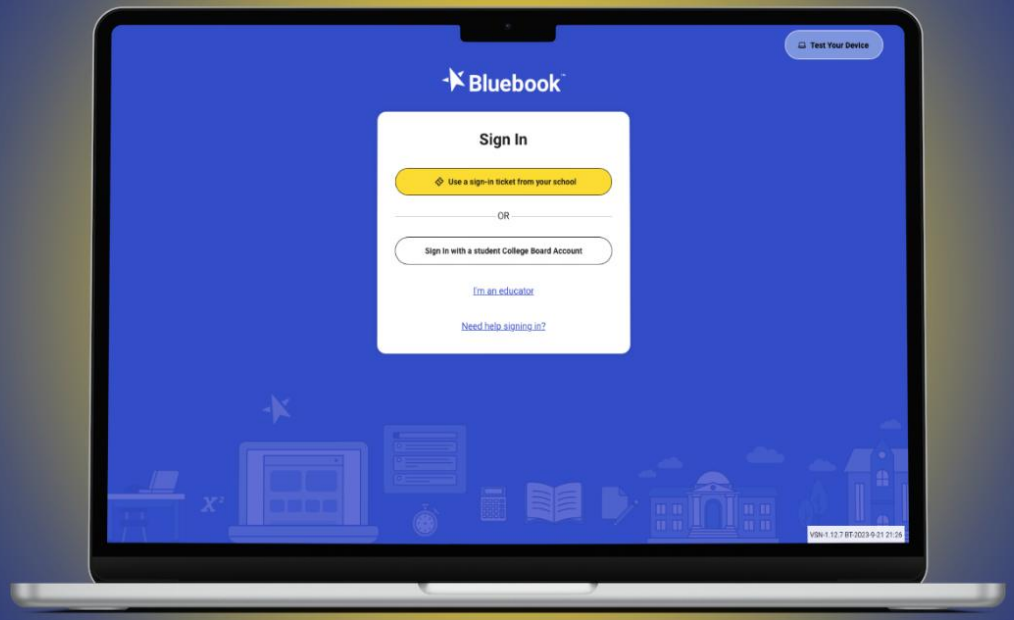
The Exam is Fully Digital

SECTION 4

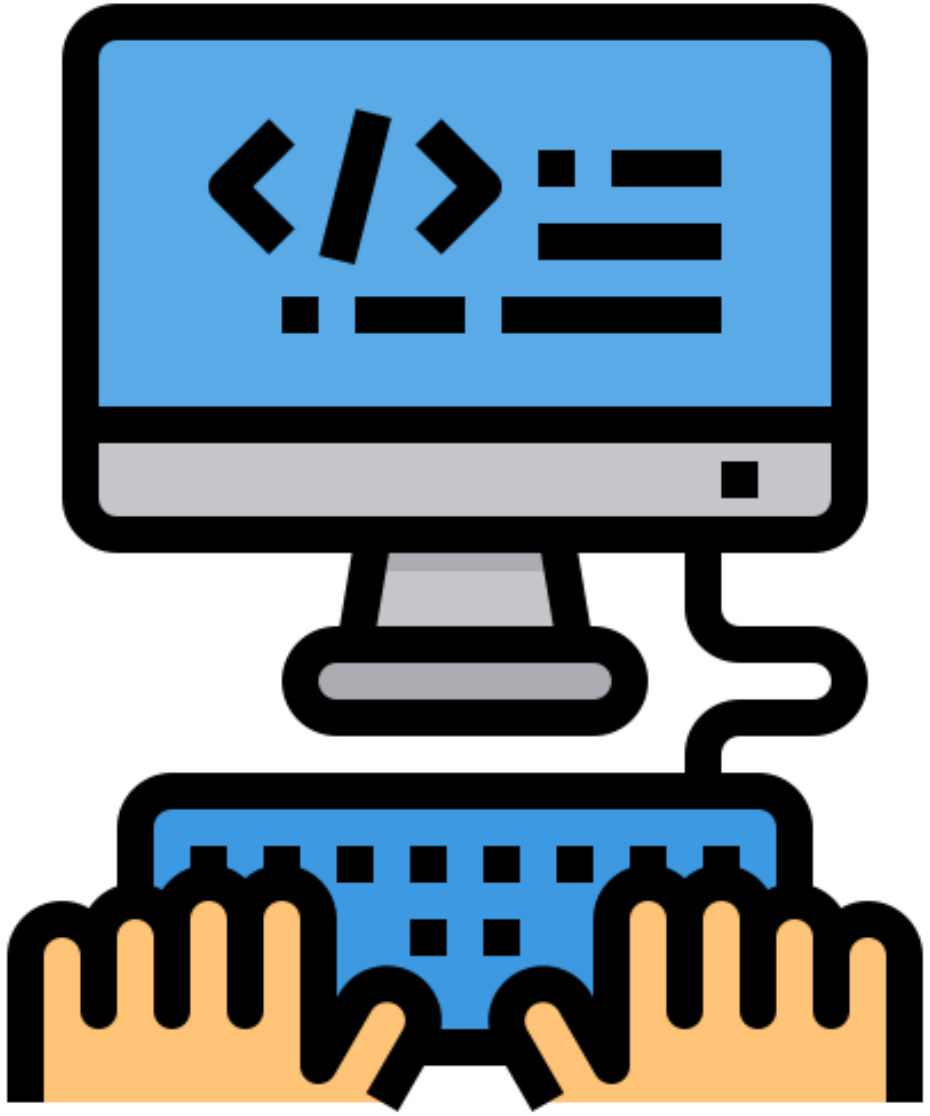
Machine

- College board Bluebook testing app.
- Installation of the on your machine or school's machine
- Windows, MacOS, iPad, Tablets, Chromebooks are OK
- No smartphone
- Extra paper available
- Bluebook annotation available
- Non-adaptive
- No penalty on guessing (MCQ)
- Rollbacks allowed

Out With The Old, In With The New



Meet College Board's "Bluebook"

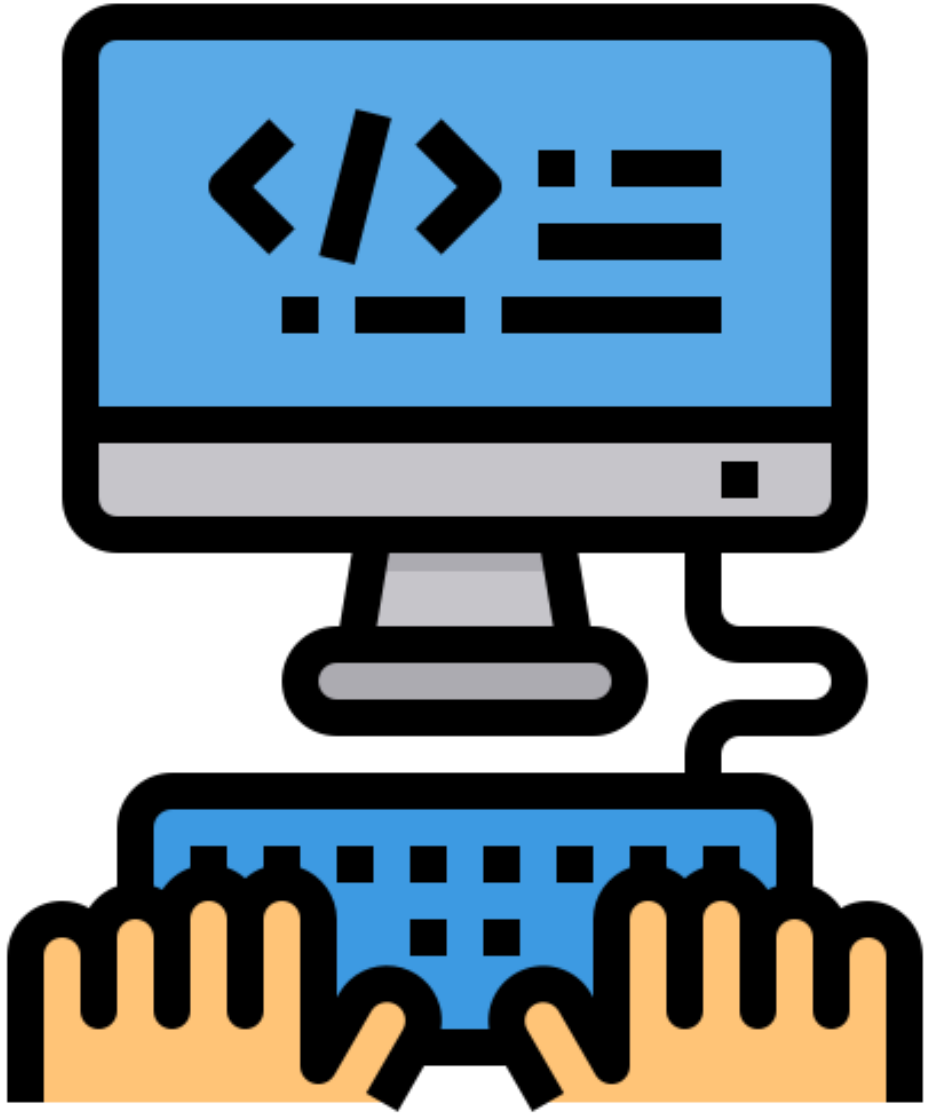


Multiple Choice Section

SECTION 5

MCQ Strategies

- Process of Elimination
- No penalty on guessing
- Be Strategic About Long Questions
- Don’t turn a Question into a crusade! (Roughly 2 mins per question)
- Review (Two-pass system)



Free Response Section

SECTION 7

FRQ Strategies

- Write Java. No other code
- Don’t comment on Your Code
- Don’t write More than one Solution to a problem
- Don’t leave any problem blank
- Keep it Simple. If it is too long (more than 30 lines), very likely it is wrong.
- Write standard solutions
- If the Pseudocode for an Algorithm is Given, Use it!

FRQ Strategies

- If Part (a) is given, Part (b) must call that function. (Same for other functions given, do not re-write new functions)
- Answer part (b) even if you can't answer part (a)
- Don't make easy-to-avoid mistakes
- Use the Scratch paper
- Don't do more work than you need to
- Understand the problem first. Do not rush to write code
- Keep track of time (15-25 mins per problem)

Question 1: Methods and Control Structures (7 pts)

- This tests your ability to call methods, write code for methods, and write code to create objects of a class.
- The question may include String manipulation, or expressions that used div (/) and mod (%), or sequences of if … else statements.

Question 2: Class Writing (7 pts)

- This tests your ability to write code for a class or subclass according to various specifications
- Brush up on writing constructors, or use of super, and overridden and inherited methods.

Question 3: Data Analysis and ArrayList Manipulation (5 pts)

- This tests your ability to create, traverse, and manipulate elements in a 1D array or ArrayList of objects.
- Review the rules for using enhanced for loops in traversals, and the specific methods of ArrayList for manipulating lists of objects.

Question 4: 2D Array Manipulation (6 pts)

- This tests your ability to create, traverse, and manipulate elements in a 2D array of objects.
- Brush up on row-by-row traversals, using the fact that a 2D array is an array of 1D arrays.
- Review the rules for using enhanced for loops in 2D array traversals.